

CURRENCY CONVERTER

PROJECT REPORT

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Course
Introduction to Problem Solving and Programming
Platform: VITyarthi – Build Your Own Project

Academic Project Report

1. Introduction

Currency conversion is a common requirement to a certain group of people who have to deal with prices, expenses, travel, international transactions or academic calculations involving different countries and currencies on a daily or regular basis. Exchange rates can change over time, they fluctuate due to supply and demand, economic conditions, wars, inflation, interest rates, central bank policies, etc. Therefore, using a fixed conversion table without checking current rates based on available data can produce an outdated conversion.

My project presents a Python-based Currency Converter application that gets the current or most updated exchange-rate data from ExchangeRate-API. It allows the user to convert an amount from one available and valid currency to another currency. The application provides two ways to interact and get the output through the program: a command-line interface (CLI) for direct terminal execution on a terminal interface and a Graphical user interface (GUI) developed using Tkinter, which provides a graphical/visual interface. The user can select any of the following or both for the desired output.

The application uses JSON data returned by the API. The input rates given by the user uses USD as the base currency. The conversion logic firstly changes the non-USD source amount into its USD equivalent and then converts that USD amount into the required output currency.

- The final result is rounded off to two decimal places to simplify it.
- The project implements and includes input and currency validation. If the external API cannot be reached due to poor connection or wifi disconnection, the application falls back to system defined rates for USD, INR, and EUR so that the program can still run offline and convert the amount.

1.1 Objectives

- Develop a simple currency conversion application in Python language.
- Get exchange-rate datas from an external API.
- Allow conversion between available currencies while using USD as the base currency for conversion.
- Provide both terminal-based and graphical user interface options.
- Handle invalid inputs for amount and unavailable/non-existent currency codes.
- Provide predefined fallback rates when calling the API results in failure or it is unavailable.

1.2 Technologies Used

- Python 3.14.7
- Tkinter
- Requests
- JSON
- ExchangeRate-API

2. Problem Statement and Proposed Solution

2.1 Problem Statement

Manual currency conversion could involve repetitive calculations and may depend on exchange-rate information that is not updated regularly based on new trends. A user may have to navigate a larger and more complex financial website even if he/she/they has/have to convert one amount from one currency to another currency.

This project addresses these problem by providing a simple and non-complex Python application in which the user enters an amount, chooses the source currency and target currency i.e., the currency he/she wants to make the conversion in and receives the result calculated by the program. The application can be used through the terminal (CLI) or through a Tkinter-based graphical interface (GUI).

2.2 Proposed Solution

The proposed solution includes the application of ExchangeRate-API with a Python application. When the program starts, it attempts to gather the latest available rates. The application then processes the user's input and performs the conversion using the inputted rates. The GUI provides dropdown menu options for currency selection, while the CLI asks for the same information through text prompts through the terminal.

The program also contains a small, simple and predefined fallback rate set. If the API request fails due to unstable connection or call, the program reports the network problem and uses those fallback values instead of stopping immediately or showing errors.

2.3 Scope

- External integration with ExchangeRate-API.
- Processing of JSON exchange-rate data.
- Conversion between currencies using USD as the base.
- CLI execution without requiring the GUI.
- Tkinter GUI as an alternative visual interface.
- Validation of empty or non-numeric amounts.
- Validation of currency availability.
- Fallback conversion using predefined rates when the API is unavailable.

2.4 Target Users

- International travellers
- Students and researchers
- Small business users
- General users

3. Functional Requirements

3.1 Exchange-Rate Retrieval

The application sends an HTTP request to ExchangeRate-API to get the latest exchange rates and reads the JSON response. The rates returned by the API are stored for use during the program execution.

3.2 Currency Conversion

The user provides an amount and selects a source and target currency. The conversion logic uses the available rates to calculate the target amount. If the source currency is not USD, the source amount is first converted to its USD equivalent and then into the target currency.

3.3 User Interaction

The project supports two interfaces. In CLI mode, the program collects the amount and currency codes through terminal input. In GUI mode, the user enters the amount in a text field and selects currencies from dropdown menus before pressing the conversion button.

3.4 Input Validation

- Checks if the amount field is empty.
- Checks if the entered amount is numeric.
- Checks whether the source currency exists in the given rates.
- Checks whether the target currency exists in the given rates.

3.5 Fallback Operation

If the API request fails, predefined fallback rates are used. The fallback set contains USD, INR, and EUR values. This can happen if there is no internet connection or the API fails to be accessed.

4. Non-Functional Requirements

- Usability: keeps the conversion process simple.
- Reliability: network failures are handled with predefined fallback rates.
- Error handling: invalid input is handled through messages.
- Maintainability: functions and the Currency_Converter class organize the main operations into functions and classes.
- Portability: the application uses Python and commonly available Python components.

5. System Architecture and Design

The application follows a simple flow in which the user interacts with either the CLI or the Tkinter GUI. Both interfaces use the same basic conversion concept: obtain rates, validate the selected currencies and amount, perform the conversion, and display the result.

5.1 Main Components

- User Interface: accepts amount and currency selections and displays the result.
- Currency Converter: stores rate information and performs conversion.
- API Communication: sends an HTTP request using Requests and reads JSON data.
- Validation: checks amount and currency selections before conversion.
- Fallback Data: provides a small predefined set of rates after an API failure.

5.2 Data Flow

1. Application starts and attempts to obtain exchange-rate data.
2. API returns JSON data containing currency rates.
3. The application stores the returned rates in memory.
4. The user enters an amount and chooses source and target currencies.
5. The application validates the input.
6. The conversion formula is applied.
7. The result is rounded to two decimal places and displayed.

5.3 Data Representation

The API response is JSON. The program extracts the 'rates' object from the response and stores it as a Python dictionary. Currency codes act as keys, while their corresponding exchange-rate values are used during calculation.

6. Design of the program

6.1 Use Case

The main actor is the User. The user can launch the application, enter an amount, select source and target currencies, request a conversion, and view the result. The system also communicates with ExchangeRate-API to obtain exchange-rate data.

6.2 Workflow

Start → Launch Application → Fetch Rates → API Available? yes/no

1. *Yes → Store Rates → Get User Input → Validate → Convert → Display Result → End*
2. *No → Use Fallback Rates → Get User Input → Validate → Convert → Display Result → End*

6.3 Workflow Explanation

The workflow begins when the application is launched. It attempts to gather rates from the API. If the request succeeds, the received rates are used. If the request fails, the fallback rate dictionary is used. The user then enters the amount and selects the source and target currencies. After verification, the conversion is performed and the result is displayed.

7. Sequence and Conversion Logic

7.1 Sequence

The sequence of interaction is slightly different depending on whether CLI or GUI mode is used, but the underlying conversion process remains the same.

*User → Interface → CurrencyConverter → API
API → CurrencyConverter: JSON rates
User → Interface: amount + source + target
Interface → CurrencyConverter: conversion request
CurrencyConverter → Interface: calculated result
Interface → User: display result*

7.2 Conversion Algorithm

- Receive the amount, source currency and target currency.
- Check that both currencies are present in the available rates.
- If the source currency is not USD, divide the amount by the source currency rate.
- Multiply the resulting USD amount by the target currency rate.
- Round the result to two decimal places.
- Display the result.

7.3 Conversion Formula

The API uses USD as the base currency. Therefore, when the source is not USD:

$$USD\ Amount = Amount / Source\ Currency\ Rate$$

The target amount is then calculated as:

$$Target\ Amount = USD\ Amount \times Target\ Currency\ Rate$$

7.4 Example

For example, if user enters 100 USD as the input and selects INR as the target currency, the application multiplies the USD amount by the INR rate received from the API. The exact result depends on the exchange-rate data available at the time of the request.

8. Implementation Details

8.1 API Request

The application sends a GET request to the ExchangeRate-API endpoint using the requests library. A timeout is used so that a network problem does not leave the request waiting infinitely. The response is checked and converted from JSON into Python data.

8.2 CurrencyConverter Class

The CurrencyConverter class is responsible for maintaining rate information and performing conversion. During initialization, the class creates its rate dictionary and attempts to fetch the available rates.

8.3 CLI Implementation

CLI mode is the default execution mode. The user is prompted for an amount, source currency and target currency. The program handles an empty amount, non-numeric input and unavailable currency codes with messages.

8.4 GUI Implementation

The GUI is implemented using Tkinter. It contains an amount input field, source and target currency dropdown menus, a conversion button, and a result display.

8.5 Execution Commands

CLI mode:

```
python Currency_Converter.py
```

GUI mode:

```
python Currency_Converter.py --gui
```

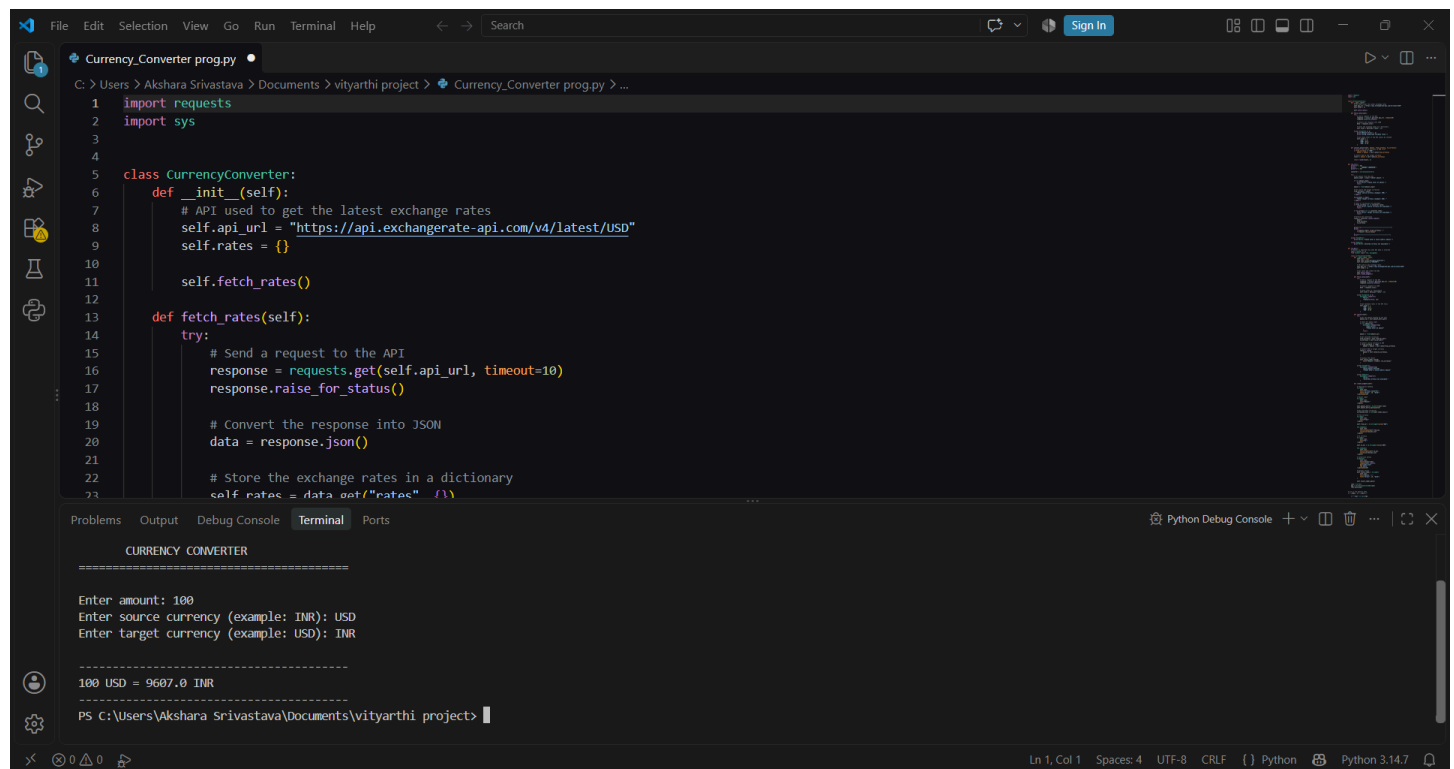
8.6 Dependency Installation

```
python -m pip install requests
```

9. Screenshots and Results

This includes screenshots demonstrating both interfaces and important cases.

9.1 CLI Successful Conversion



```
Currency_Converter prog.py
C:\Users\Akshara Srivastava\Documents\vityarthi project> Currency_Converter prog.py > ...

1 import requests
2 import sys
3
4
5 class CurrencyConverter:
6     def __init__(self):
7         # API used to get the latest exchange rates
8         self.api_url = "https://api.exchangerate-api.com/v4/latest/USD"
9         self.rates = {}
10
11     def fetch_rates(self):
12
13     def fetch_rates(self):
14     try:
15         # Send a request to the API
16         response = requests.get(self.api_url, timeout=10)
17         response.raise_for_status()
18
19         # Convert the response into JSON
20         data = response.json()
21
22         # Store the exchange rates in a dictionary
23         self.rates = data.get("rates") {}
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10. Testing and Validation

Test was performed by running the application through the terminal and through the Tkinter interface GUI. The tests focus on normal conversion, invalid input, unavailable currencies and API/network failure handling.

10.1 Standard Conversion Test

Run the application.
Enter a valid amount such as 100.
Select valid source and target currencies.
Perform the conversion.
Expected result: the calculated target amount is displayed.

10.2 Empty Amount Test

Leave the amount field empty.
Attempt to perform the conversion.
Expected result: the application displays an input warning/error instead of attempting a numerical conversion.

10.3 Non-Numeric Amount Test

Enter a value such as 'abc' in the amount field.
Attempt to perform the conversion.
Expected result: the application reports that a valid numeric amount is required.

10.4 Currency Availability Test

Enter a currency code that is not present in the available rates.
Expected result: the application reports that the selected currency is not available.

10.5 Network/Fallback Test

Run the application when the external API cannot be reached.
Expected result: the application handles the network failure and uses predefined fallback rates for USD, INR and EUR.

10.6 Testing Summary

Test Case	Condition	Expected Result
Normal conversion	Valid amount and currencies	Converted amount displayed
Empty amount	No amount entered	Input warning/error displayed
Non-numeric amount	Amount = abc	Numeric input error displayed
Invalid currency	Unavailable currency code	Currency availability error displayed
API failure	API unreachable	Fallback rates used

11. Challenges Faced and Solutions

11.1 Working with an External API

One of the challenges were handling data received from an external service. The application had to request the data, interpret the JSON response and store it in the exchange-rate dictionary. A timeout and exception handling were included so that a failed request can be handled.

11.2 Handling User Input

User input cannot always be assumed to be valid. Empty input, non-numeric values and unavailable currency codes can cause conversion problems. The application therefore checks the input before performing the calculation and displays appropriate messages.

11.3 Connecting GUI and Logic

The GUI needs to obtain values from its input fields and pass them to the conversion logic. Tkinter widgets such as Entry, Combobox and Button are used to collect input and trigger the conversion operation.

11.4 Network Failure Handling

The API depends on network connectivity. To make the application more resilient during a failed request, predefined fallback rates are used. This does not provide live market data during a network failure, but it allows the application to continue demonstrating basic conversion functionality.

12. Learnings and Key Takeaways

1. Learn how to make HTTP requests from Python using request library/module.
2. Learn how to read JSON data and extract values from a Python dictionary.
3. Understand how exchange rates can be used for cross-currency calculations.
4. Practise input validation and exception handling.
5. Learn how to build a simple desktop GUI using Tkinter.
6. Understand the differences between terminal-based and graphical user interface.
7. Learn the importance of handling external-service failures.

13. Limitations

- Currency conversion depends on ExchangeRate-API availability and an internet connection.
- Fallback rates are fixed predefined values and may not reflect current market rates.
- The application does not store previous conversion history.
- The project does not provide previous exchange-rate data.
- The application is a conversion utility and not a financial trading system.

14. Future Enhancements

- Add conversion history using suitable local storage.
- Add graphical charts for historical exchange-rate trends.
- Add more detailed input validation and user feedback.
- Improve GUI customization and accessibility.
- Add additional financial conversion features.
- Add optional support for other asset types where suitable APIs are available.

15. Conclusion

The Currency Converter project demonstrates the use of Python programming concepts to solve a practical problem. The project combines an external API, JSON data processing, currency conversion logic, input validation, exception handling, a command-line interface and a Tkinter graphical interface.

The project also demonstrates the ability to use fallback rates in case of an API failure when an external service is unavailable. Through the project, I gained practical experience working with API integration, user input handling, GUI development, program applications and program organization.

16. References

Python Documentation – <https://docs.python.org/3/>

Tkinter Reference – <https://tkdocs.com/>

ExchangeRate-API Documentation – <https://www.exchangerate-api.com/docs>

VITyarthi – Build Your Own Project guidelines and course project requirements.

THE END